Tactile Robotics · Lab Session

From Touch to Simulation

Taxels · DEXH13 · MuJoCo — a guided walkthrough

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Gesture Teleoperation Demo

Watch first. One question in your head while it runs:



WHAT TO WATCH

How the hand follows the gesture

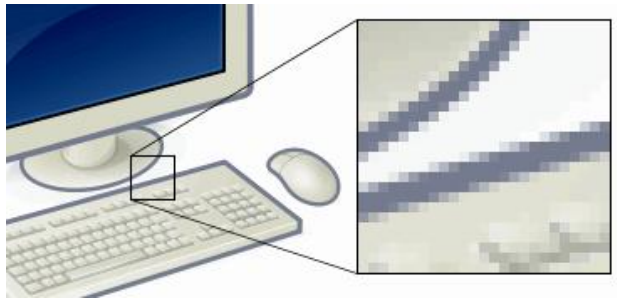
The moment it first touches the object

When grip changes — tighten, loosen, slip

What Changes When the Robot Touches an Object?

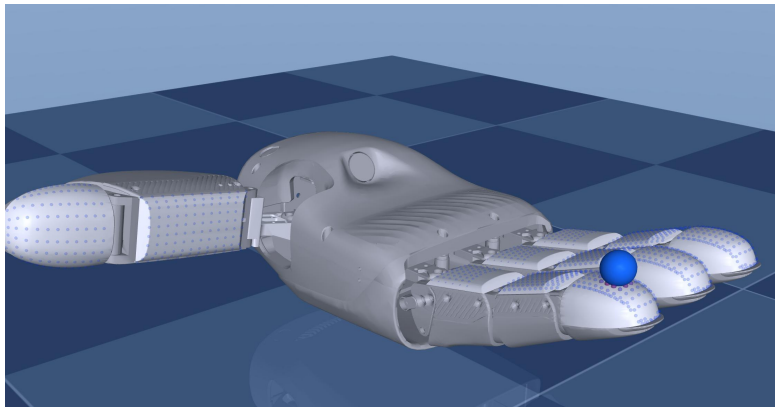
Camera / vision tells us	Tactile tells us
Where the hand and object are	Pressure distribution on contact
Shape, orientation, movement	How hard each finger presses
Often not how hard it presses	Slip, squeeze, in real time

Pixel and Taxel



Pixel

The smallest unit of a digital image. Millions of them form what you see on a screen.



Taxel

The smallest unit of a tactile image. Hundreds of them form what a skin feels.

HOW ONE TAXEL WORKS

A taxel converts contact into an electrical signal. Common principles:
Capacitive — pressure changes plate distance → capacitance.
Piezoresistive — material resistance changes under force.
Optical — elastomer deformation shifts reflected light.

DEXH-13 Tactile Sensors

TAXELS

1,140

PATCH TYPES

5



Tactile Monitoring Interface

WHAT YOU SEE

Live pressure field across every patch

Intensity mapped to color (blue → red)

Where force concentrates, where it slips

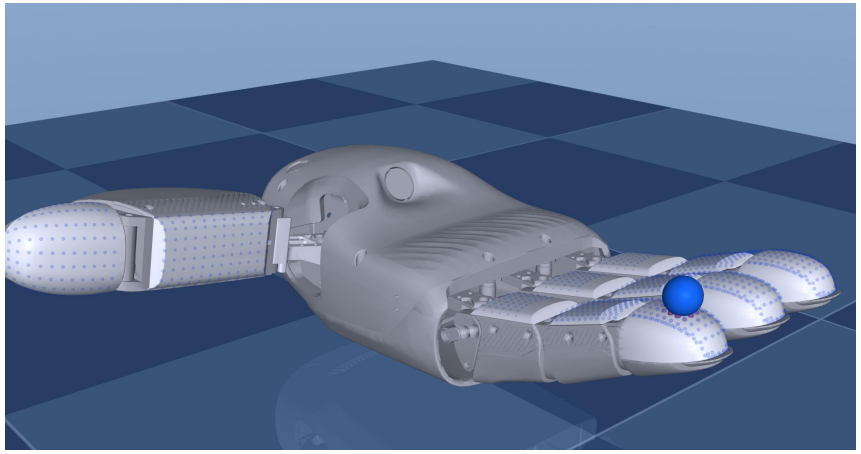
Now it's your turn — grab the hand, and watch the heatmap.

1 Hold the bottle. Where is the pressure brightest on the heatmap?

2 Squeeze tighter, then relax. How does the color change?

3 Try to make it slip. What does the heatmap show at the moment it slips?

Why Use Simulation?



Real hardware is slow, fragile, and expensive. Touch is no exception.

Safety

Test aggressive grasps and failure cases without damaging fingers or objects.

Cost & scale

Thousands of trials in parallel, no wear on real sensors.

Repeatability

Identical initial conditions every run; controlled contact geometry.

Data & learning

Generate labeled tactile data for policies before transfer to real robots.

MuJoCo Installation

We use MuJoCo as our physics engine. Let's get it installed.

```
# create an environment (recommended) / conda
python -m venv venv
source venv/bin/activate      # Windows: venv\Scripts\activate
pip install mujoco
```

Installation Check

Run this to confirm MuJoCo works on your machine.

RUN

```
python -c "import mujoco; print(mujoco.__version__)"
```

EXPECTED

A version number prints, e.g. 3.x.x — no error. Then open the viewer to see a scene render.

Wrap-up

Seen a tactile teleoperation demo, and what touch adds beyond vision.

Held DEXH13 — 1,140 taxels, 5 patch types, a live heatmap.

Set up MuJoCo on your own machine — next: taxels in simulation.

Next time: build a taxel in MuJoCo, calibrate it against the real sensor, and start the sim-to-real loop.

Questions?